Machine Learning Viva

Most Important Questions & Answers | All 5 Units

UNIT 1: Introduction to Machine Learning

Q: What is Machine Learning? Define it formally.

■ ML is a field of AI that gives computers the ability to learn from data without being explicitly programmed. Tom Mitchell's formal definition: 'A computer program is said to learn from experience E with respect to task T and performance measure P, if its performance on T, as measured by P, improves with experience E.'

■ *Concept: Three key elements: Task (T), Experience (E), Performance (P). Example: Email spam filter — T=classify spam, E=labelled emails, P=accuracy.*

Q: What are the three main types of Machine Learning?

■ 1. Supervised Learning: Model trained on labelled data (input-output pairs). Examples: Classification, Regression. 2. Unsupervised Learning: Model finds hidden patterns in unlabelled data. Examples: Clustering, PCA. 3. Reinforcement Learning: Agent learns by interacting with environment and receiving rewards/penalties.

■ *Concept: Supervised = teacher present; Unsupervised = no teacher; Reinforcement = learn from consequences.*

Q: What is the difference between Classification and Regression?

■ Classification predicts discrete/categorical output (e.g., spam/not spam, disease yes/no). Regression predicts continuous numerical output (e.g., house price, temperature). Both are supervised learning tasks.

■ *Concept: Classification → discrete labels. Regression → continuous values.*

Q: What are the applications of Machine Learning?

■ Image recognition, speech recognition, natural language processing (NLP), recommendation systems (Netflix, Amazon), fraud detection, medical diagnosis, autonomous vehicles, spam filtering, stock price prediction.

■ *Concept: Be ready to give at least 4-5 real-world examples with the type of ML used (supervised/unsupervised).*

Q: What is Reinforcement Learning? Give an example.

■ RL is a type of ML where an agent learns to make decisions by interacting with an environment. It receives rewards for good actions and penalties for bad ones, aiming to maximise cumulative reward. Example: Training a robot to walk, AlphaGo playing chess.

■ *Concept: Key terms: Agent, Environment, State, Action, Reward, Policy.*

Q: What is the difference between a model, algorithm, and training in ML?

■ Algorithm is the procedure/method (e.g., Decision Tree). Training is the process of applying the algorithm to data to produce a Model. The Model is the trained output that makes predictions on new data.

■ *Concept: Algorithm + Data → Training → Model → Predictions.*

UNIT 2: Preprocessing

Q: What is Feature Scaling? Why is it needed?

■ Feature scaling normalises the range of features so no single feature dominates due to large magnitude. It's needed for algorithms sensitive to scale like KNN, SVM, Gradient Descent, PCA. Methods: Min-Max Normalization (scales to [0,1]) and Standardization (z-score, mean=0, std=1).

■ *Concept: Min-Max: x' = (x - xmin)/(xmax - xmin). Standardization: x' = (x - mean)/std. Decision Trees DON'T need scaling.*

Q: What is the difference between Normalization and Standardization?

■ Normalization (Min-Max Scaling) rescales values to [0,1] range. Best when distribution is not Gaussian. Standardization (Z-score) rescales to mean=0, std=1. Best when data follows Gaussian distribution. Standardization is less affected by outliers.

■ *Concept: Use Normalization for image pixels (bounded). Use Standardization for most ML algorithms.*

Q: What are Feature Selection methods?

■ Feature selection removes irrelevant/redundant features to reduce dimensionality. Methods: 1. Filter Methods (use statistical measures, e.g., correlation, chi-square). 2. Wrapper Methods (use model performance, e.g., RFE). 3. Embedded Methods (built into algorithm, e.g., Lasso regularisation).

■ *Concept: Reduces overfitting, improves accuracy, reduces training time.*

Q: What is PCA (Principal Component Analysis)?

■ PCA is a dimensionality reduction technique that transforms features into a new set of uncorrelated variables called Principal Components. These are ordered by the amount of variance they capture. Steps: Standardise data → Compute covariance matrix → Find eigenvectors/eigenvalues → Select top k components → Transform data.

■ *Concept: PCA is unsupervised. It reduces dimensions while preserving maximum variance. Used for visualisation and removing multicollinearity.*

Q: What is Dimensionality Reduction and why is it important?

■ Dimensionality reduction reduces the number of input features. It helps in: (1) Removing the curse of dimensionality, (2) Reducing overfitting, (3) Faster training, (4) Better visualisation. Techniques: PCA (unsupervised), LDA (supervised), t-SNE (for visualisation).

■ *Concept: Curse of dimensionality: as features increase, data becomes sparse and models perform worse.*

Q: What is the curse of dimensionality?

■ As the number of dimensions (features) increases, the volume of the feature space grows exponentially. Data becomes increasingly sparse, making it hard for algorithms to find meaningful patterns. Distance metrics become less meaningful in high dimensions.

■ *Concept: Solution: Dimensionality reduction (PCA), feature selection, regularisation.*

UNIT 3: Regression

Q: What is Linear Regression? Write its hypothesis.

■ Linear Regression models the relationship between a dependent variable y and one or more independent variables x by fitting a linear equation. Simple LR: h(x) = theta0 + theta1*x. Multiple LR: h(x) = theta0 + theta1*x1 + theta2*x2 + ... + thetaN*xN. The goal is to find parameters theta that minimise the cost function.

■ *Concept: Assumes: linearity, independence of errors, homoscedasticity, normality of residuals.*

Q: What is the Cost Function in Linear Regression?

■ The cost function (Mean Squared Error) measures how far predictions are from actual values. J(theta) = (1/2m) * sum[(h(xi) - yi)^2] where m is number of training examples. We minimise J(theta) to find best-fit parameters. Factor 1/2 makes derivative cleaner.

■ *Concept: MSE penalises large errors more (squared). Always convex for linear regression — one global minimum.*

Q: Explain Gradient Descent. What are its types?

■ Gradient Descent is an optimisation algorithm to minimise cost function. Update rule: theta_j := theta_j - alpha * (dJ/d theta_j). Types: (1) Batch GD: uses entire training set per update — slow but stable. (2) Stochastic GD (SGD): uses one sample per update — fast but noisy. (3) Mini-batch GD: uses small batch — balance of both. Learning rate alpha controls step size.

■ *Concept: Too large alpha: overshoots, diverges. Too small alpha: very slow convergence. Optimal alpha converges smoothly.*

Q: What is Overfitting and Underfitting?

■ Overfitting: Model learns training data too well including noise, performs poorly on new data (high variance). Underfitting: Model is too simple, fails to capture underlying pattern (high bias). Ideal: balanced bias-variance tradeoff with good generalisation.

■ *Concept: Overfitting solutions: more data, regularisation (L1/L2), dropout, cross-validation, simpler model.*

Q: What is Regularisation? Explain L1 and L2.

■ Regularisation adds a penalty term to the cost function to prevent overfitting by discouraging large weights. L1 (Lasso): adds sum of |theta_j|. Produces sparse models (some weights become exactly 0 — feature selection). L2 (Ridge): adds sum of theta_j^2. Shrinks weights towards zero but rarely exactly 0. Elastic Net combines both.

■ *Concept: L1 → sparse (feature selection). L2 → small weights. Lambda controls regularisation strength.*

Q: What are Regression Evaluation Metrics?

■ 1. MAE (Mean Absolute Error): average of |y - y_hat|. Robust to outliers. 2. MSE (Mean Squared Error): average of (y - y_hat)^2. Penalises large errors. 3. RMSE: square root of MSE, same units as y. 4. R-squared (R2): proportion of variance explained by model. R2=1 is perfect, R2=0 means model is as good as mean baseline.

■ *Concept: R2 can be negative if model is worse than baseline. RMSE is most commonly used in practice.*

Q: What is Polynomial Regression?

■ Polynomial Regression fits a polynomial curve to data by adding polynomial features (x^2, x^3, etc.). It is still linear in parameters (hence solved by linear regression) but nonlinear in features. Used when relationship between variables is nonlinear. Prone to overfitting with high-degree polynomials.

■ *Concept: Degree 1 = linear. Degree 2 = quadratic. Higher degree → risk of overfitting.*

UNIT 4: Classification

Q: What is a Decision Tree? How does it work?

■ A Decision Tree is a supervised learning algorithm that splits data into branches based on feature values to make predictions. Each internal node = test on a feature. Each branch = outcome of test. Each leaf = class label. Splitting criterion: Information Gain (ID3), Gini Index (CART), Gain Ratio (C4.5).

■ *Concept: ID3 uses entropy and information gain. CART uses Gini impurity. Prone to overfitting — use pruning.*

Q: What is Entropy and Information Gain?

■ Entropy measures impurity/uncertainty in a dataset. H(S) = -sum[p_i * log2(p_i)]. Pure node = 0, maximum impurity = 1 (for binary). Information Gain = Entropy(parent) - weighted average Entropy(children). ID3 selects the feature with maximum Information Gain at each split.

■ Concept: High entropy = high impurity. IG > 0 means the split is useful. Always pick highest IG feature.

Q: Explain Naive Bayes Classifier.

■ Naive Bayes is a probabilistic classifier based on Bayes' theorem with the 'naive' assumption that all features are conditionally independent given the class. P(C|X) proportional to P(C) * product[P(xi|C)]. It's fast, works well with text classification, spam filtering. Types: Gaussian NB, Multinomial NB, Bernoulli NB.

■ Concept: Naive assumption = features independent (often violated but still works well in practice). Laplace smoothing handles zero probabilities.

Q: What is Logistic Regression? Why can't we use Linear Regression for classification?

■ Logistic Regression is a classification algorithm that predicts probability of a binary class using the sigmoid function: sigma(z) = 1/(1+e^(-z)) where z = theta^T * x. Output is between 0 and 1. Threshold at 0.5: if output >= 0.5, predict class 1. Linear regression output is unbounded (can be <0 or >1), unsuitable for probabilities.

■ Concept: Cost function: Log Loss (cross-entropy). Optimised using Gradient Descent. Can be extended to multi-class with softmax.

Q: What is K-Nearest Neighbour (KNN)?

■ KNN is a non-parametric, lazy learning algorithm. For prediction, it finds K nearest training points to the query point (using Euclidean distance) and takes majority vote (classification) or average (regression). Choice of K matters: small K = noisy boundary, large K = smooth but may miss patterns.

■ Concept: Lazy learner = no training phase (stores all data). Sensitive to feature scaling and irrelevant features. O(n) prediction time.

Q: What is a Neural Network? Explain key components.

■ A Neural Network is inspired by the brain. Components: Input Layer (receives features), Hidden Layer(s) (learns representations via activation functions), Output Layer (produces predictions). Each neuron computes weighted sum + bias, then applies activation function (ReLU, sigmoid, tanh). Training via Backpropagation + Gradient Descent.

■ Concept: Activation functions add non-linearity. ReLU = max(0,x) most popular. Backprop computes gradients using chain rule.

Q: What is SVM (Support Vector Machine)?

■ SVM finds the optimal hyperplane that maximises the margin between two classes. Support Vectors are the data points closest to the hyperplane. Kernel trick allows SVM to handle non-linearly separable data by mapping to higher dimensions. Kernels: Linear, RBF (Gaussian), Polynomial.

■ Concept: Margin = 2/||w||. Hard margin: no misclassification. Soft margin: allows some with C parameter. C controls bias-variance tradeoff.

Q: What are Classification Evaluation Metrics?

■ 1. Accuracy = (TP+TN)/(TP+TN+FP+FN). 2. Precision = TP/(TP+FP) — of predicted positives, how many correct. 3. Recall = TP/(TP+FN) — of actual positives, how many found. 4. F1-Score = 2*(Precision*Recall)/(Precision+Recall). 5. ROC-AUC: area under ROC curve. Confusion matrix shows all TP, TN, FP, FN.

■ Concept: Use F1-score for imbalanced classes. Accuracy is misleading when classes are imbalanced. AUC=1 is perfect, AUC=0.5 is random.

Q: What is the Multilayer Perceptron (MLP)?

■ MLP is a fully connected feedforward neural network with one or more hidden layers. It uses backpropagation for training. Each layer learns increasingly abstract features. MLP can approximate any continuous function (Universal Approximation Theorem). Used for classification and regression tasks.

■ *Concept: MLP = deep learning building block. More hidden layers = deeper network = can learn complex patterns.*

UNIT 5: Clustering

Q: What is Clustering? How is it different from Classification?

■ Clustering is an unsupervised learning technique that groups similar data points together without using predefined labels. Classification is supervised (labels known during training). Clustering discovers natural groups/structure in data. Examples: customer segmentation, document grouping, anomaly detection.

■ *Concept: No labels in clustering — algorithm figures out groups itself. Evaluation is harder (no ground truth).*

Q: Explain K-Means Clustering Algorithm step by step.

■ 1. Choose K (number of clusters). 2. Randomly initialise K centroids. 3. Assign each point to nearest centroid (Euclidean distance). 4. Recompute centroids as mean of assigned points. 5. Repeat steps 3-4 until centroids don't change (convergence). Objective: minimise Within-Cluster Sum of Squares (WCSS / inertia).

■ *Concept: K-Means is sensitive to initialisation (use K-Means++) and outliers. K must be specified in advance. Assumes spherical, equal-sized clusters.*

Q: How do you choose the optimal K in K-Means?

■ Elbow Method: plot WCSS vs K values. The point where WCSS starts decreasing slowly (forming an 'elbow') is the optimal K. Silhouette Score: measures how similar a point is to its own cluster vs others. Score ranges from -1 to +1. Higher is better.

■ *Concept: There's no single perfect method. Elbow method + domain knowledge is most practical approach.*

Q: What is Hierarchical Clustering?

■ Hierarchical clustering builds a tree of clusters (dendrogram). Two types: (1) Agglomerative (bottom-up): each point starts as own cluster, merge closest pairs iteratively. (2) Divisive (top-down): all points in one cluster, split recursively. Linkage methods: Single, Complete, Average, Ward. No need to specify K in advance.

■ *Concept: Dendrogram visualises the merging process. Cut at desired level to get K clusters. Computationally O(n^3) — slow for large data.*

Q: What are Distance Metrics used in Clustering?

■ 1. Euclidean Distance: sqrt(sum(xi-yi)^2) — straight-line distance, most common. 2. Manhattan Distance: sum|xi-yi| — sum of absolute differences. 3. Cosine Similarity: measures angle between vectors, used in text. 4. Minkowski: generalisation of Euclidean and Manhattan.

■ *Concept: Euclidean for continuous data. Cosine for text/high-dim. Manhattan robust to outliers. Always scale features before computing distances.*

Q: What are the limitations of K-Means?

■ 1. Must specify K in advance. 2. Sensitive to initial centroid placement. 3. Assumes spherical, equally-sized clusters. 4. Sensitive to outliers (use K-Medoids instead). 5. Struggles with non-convex cluster shapes. 6. Only works with numerical data.

■ *Concept: Alternatives: DBSCAN for arbitrary shapes, Gaussian Mixture Models for probabilistic clustering.*

Q: What is DBSCAN and how is it different from K-Means?

■ DBSCAN (Density-Based Spatial Clustering of Applications with Noise) groups points that are closely packed and marks outliers as noise. Parameters: epsilon (neighbourhood radius) and MinPts (minimum points). Does NOT require K. Can find arbitrary-shaped clusters. Handles noise/outliers well.

■ *Concept: Core point: >= MinPts in epsilon radius. Border point: in epsilon of core but fewer neighbours. Noise point: neither. K-Means can't detect noise; DBSCAN can.*

BONUS: Frequently Asked Cross-Topic Questions

Q: What is Bias-Variance Tradeoff?

■ Bias = error from wrong assumptions (underfitting). Variance = error from sensitivity to small fluctuations (overfitting). Total Error = Bias^2 + Variance + Irreducible Noise. Simple models: high bias, low variance. Complex models: low bias, high variance. Goal: find sweet spot.

■ *Concept: Regularisation reduces variance. More data reduces variance. Feature engineering reduces bias.*

Q: What is Cross-Validation? Why is it used?

■ Cross-validation is a technique to evaluate model generalisation. K-Fold CV: split data into K folds, train on K-1 folds, test on 1 fold, repeat K times, average results. Prevents overfitting to a single train/test split. Common: 5-fold or 10-fold CV.

■ *Concept: Stratified K-Fold ensures class balance in each fold. LOOCV (K=n) is most expensive but least biased.*

Q: What is the difference between parametric and non-parametric models?

■ Parametric models assume a fixed functional form and learn a fixed number of parameters (e.g., Linear Regression, Logistic Regression, Naive Bayes). Non-parametric models make no strong assumptions about the functional form and parameters grow with data (e.g., KNN, Decision Trees, SVM with RBF kernel).

■ *Concept: Parametric: faster, less data needed, but may underfit complex data. Non-parametric: more flexible, needs more data.*

Q: What is the train-test split and why is it important?

■ Data is split into training set (to fit model), validation set (to tune hyperparameters), and test set (final unbiased evaluation). Typical splits: 70/30 or 80/20 for train/test. The test set must never be used during training to avoid data leakage.

■ *Concept: Data leakage: using test/future information during training leads to overly optimistic performance metrics.*

Best of luck for your viva, Rohit! Focus on understanding concepts, not just memorising answers. Be ready to give examples and explain the intuition behind each algorithm.